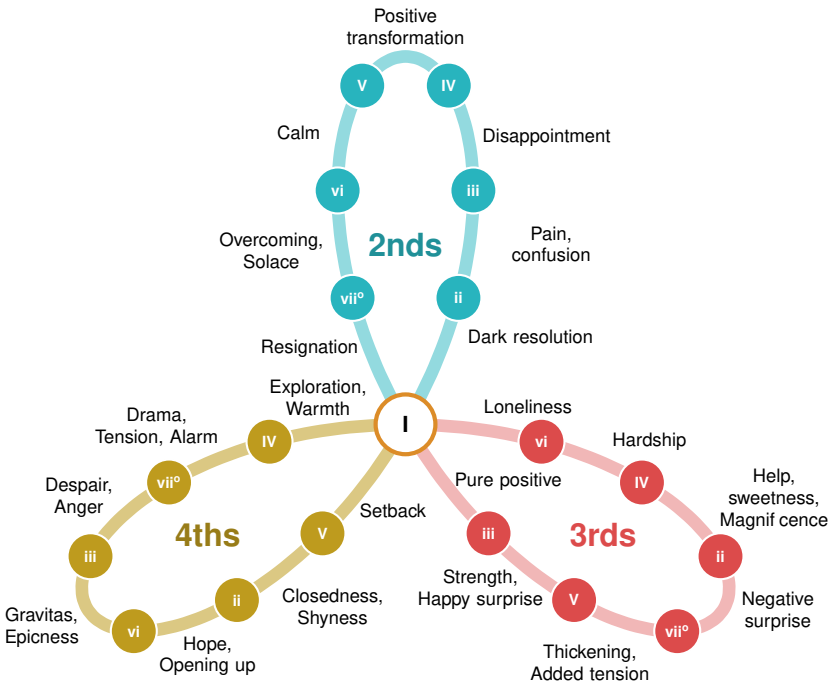


Trefoil Drill

The trefoil voicing system uses 48 asymmetric LH/RH chord pairs covering all 12 finger patterns across 6 sections.

Traversal: CCW4 → CW3 → CCW2 → CW2 → CCW3 → CW4

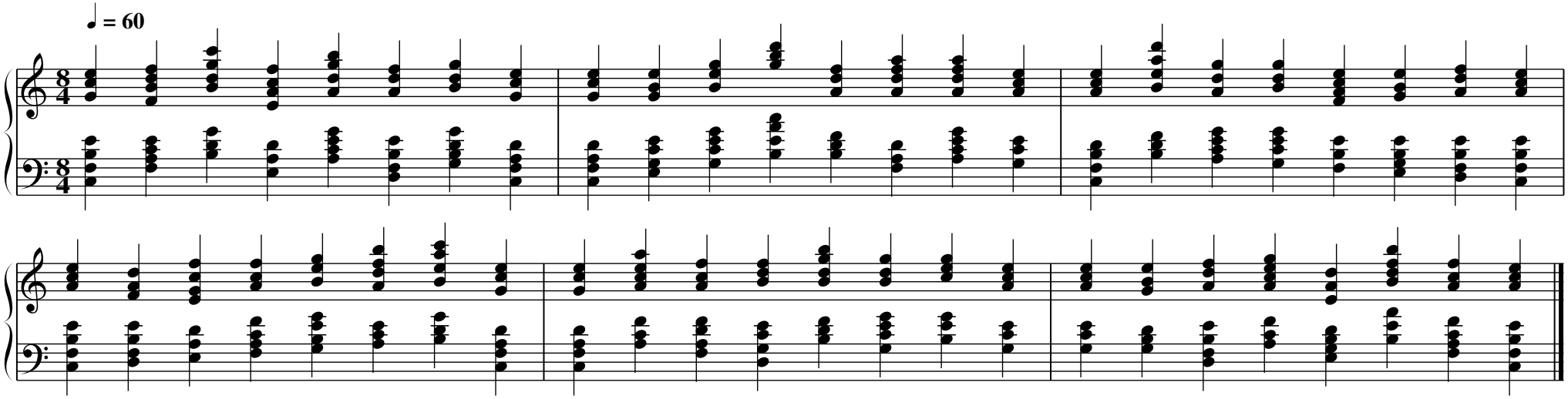
Leaves:

- **2nds** — stepwise motion
- **3rds** — skip motion
- **4ths** — circle of fifths

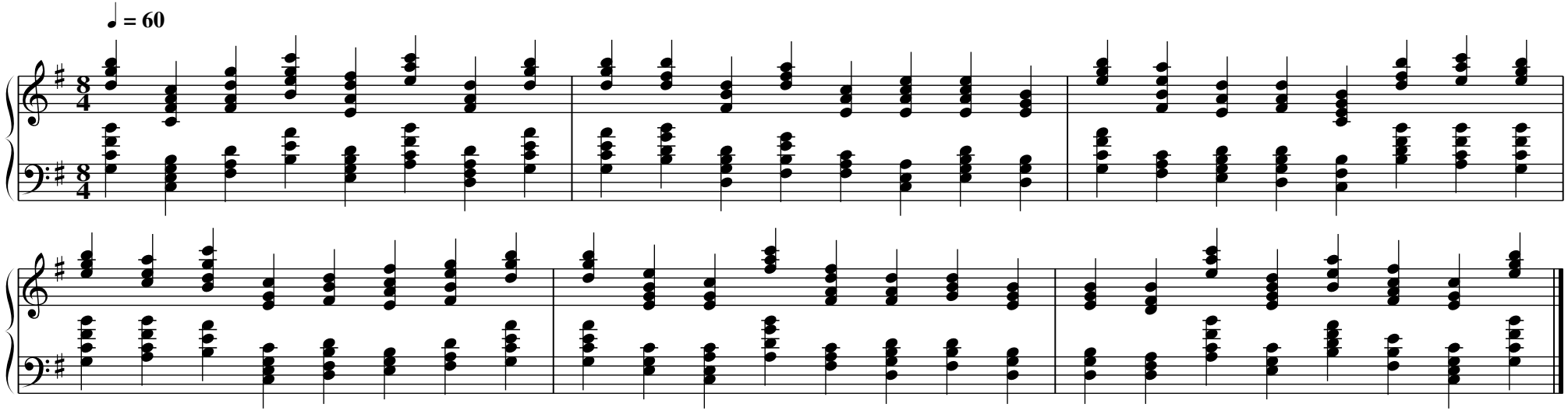
Time: 8/4 = 8 chords per measure, one section per bar. Set levers for your key, then play — no accidentals.

Keys: E♭, B♭, F, C, G, D, A

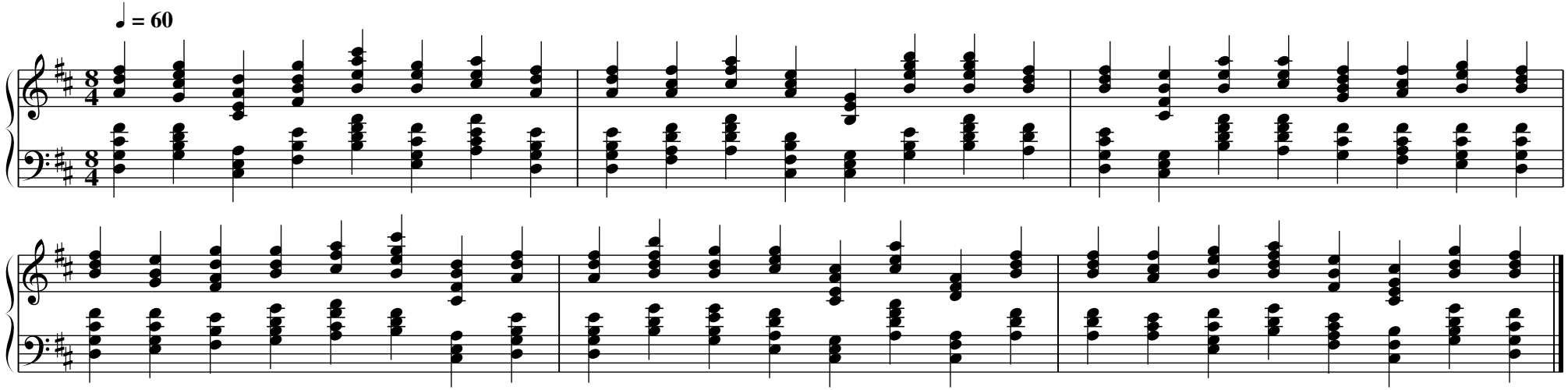
Trefoil in C



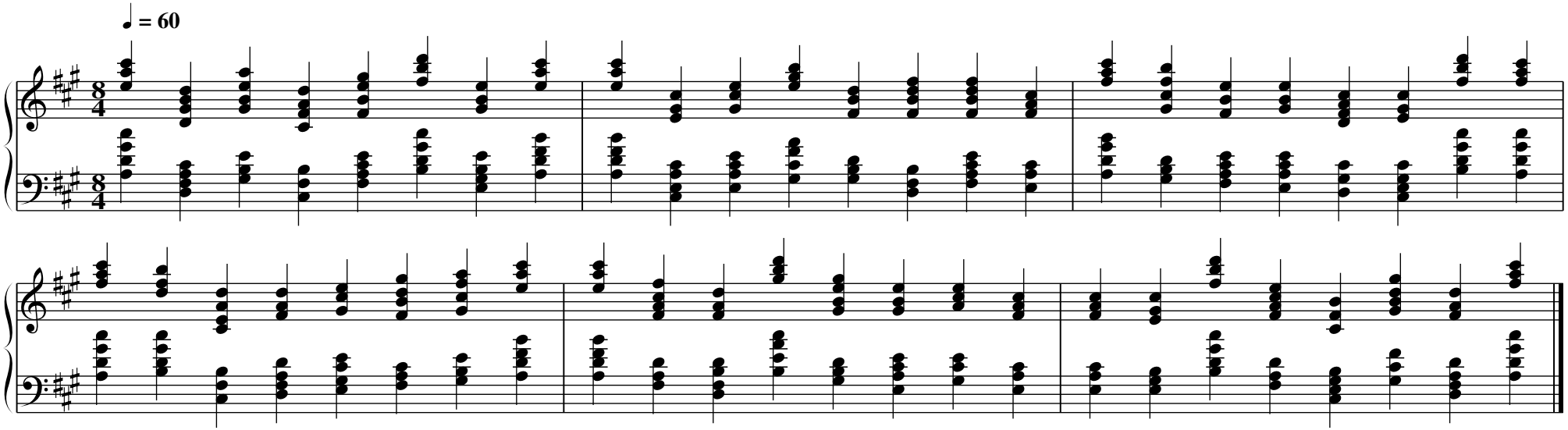
Trefoil in G



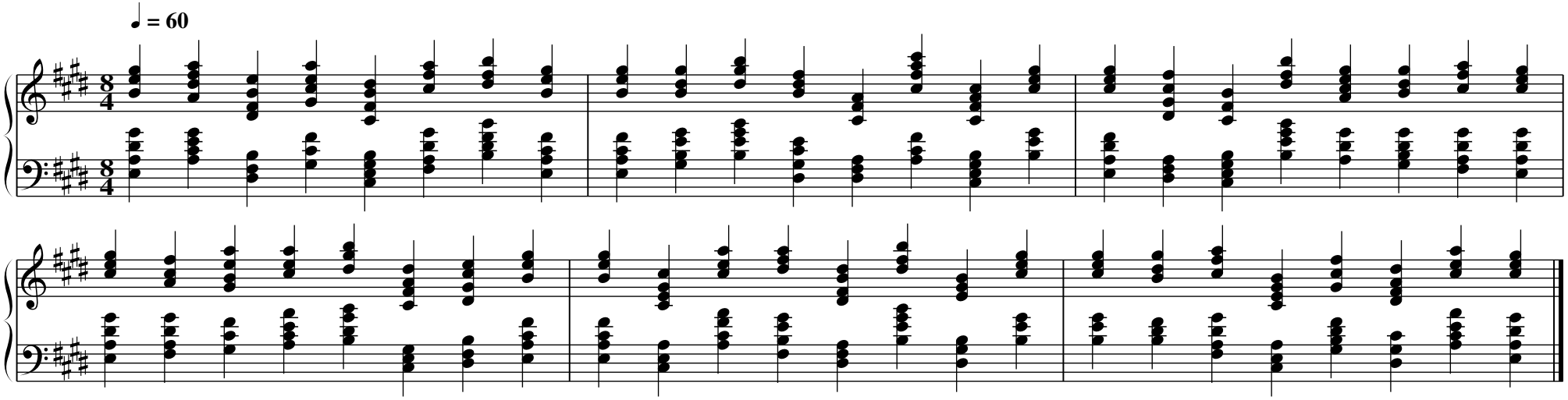
Trefoil in D



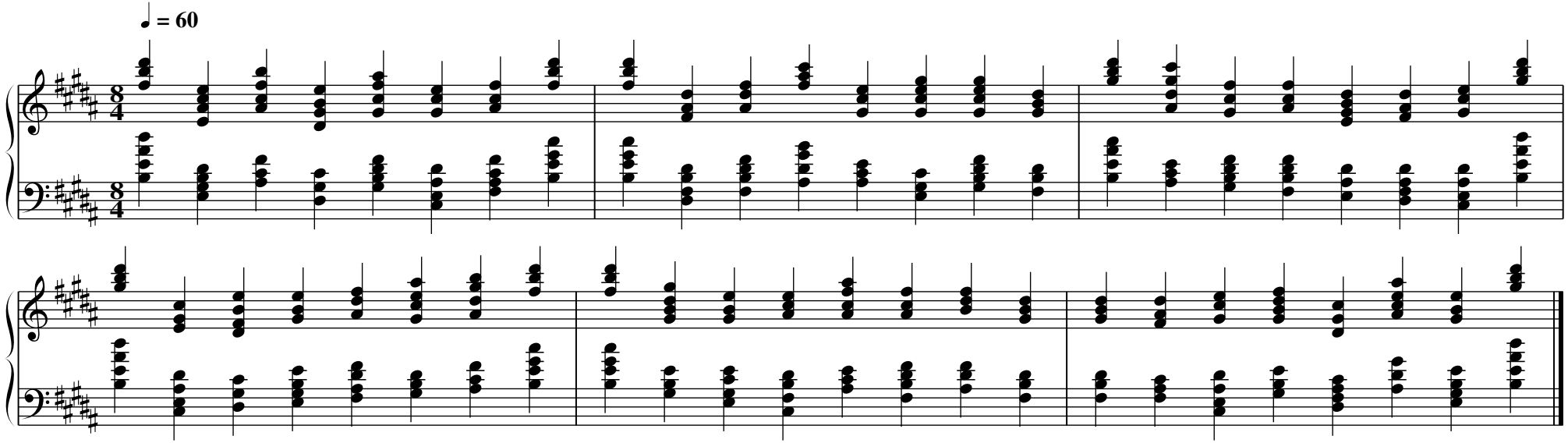
Trefoil in A



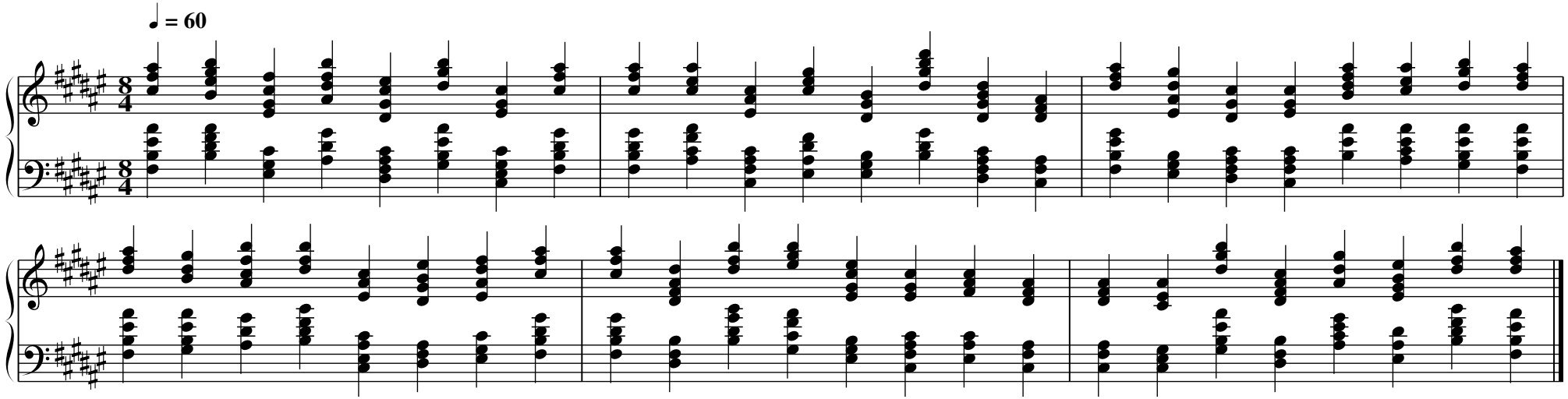
Trefoil in E



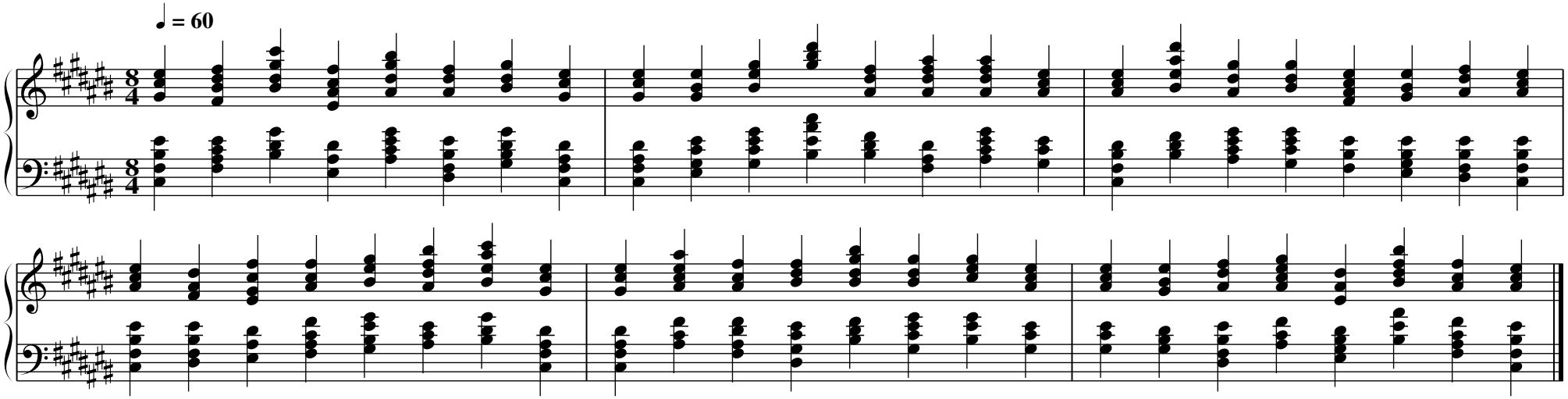
Trefoil in B



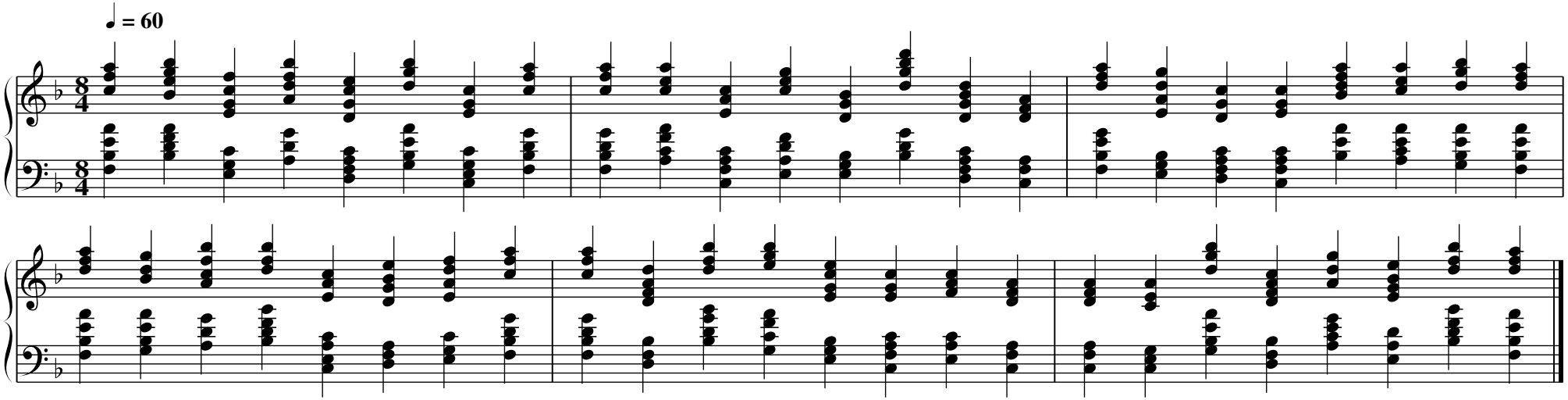
Trefoil in F#



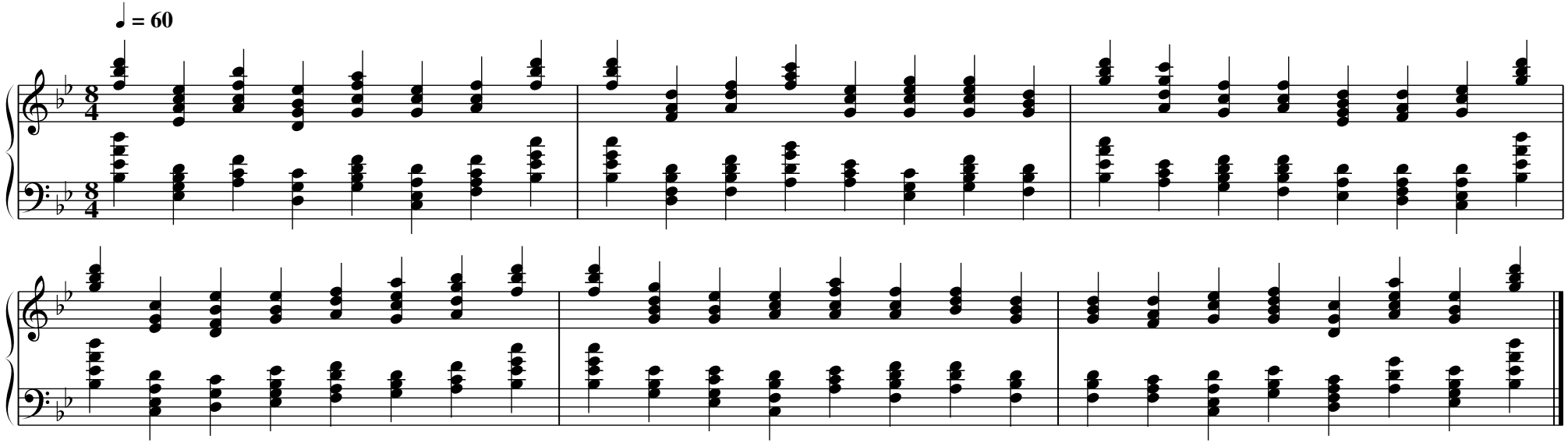
Trefoil in C#



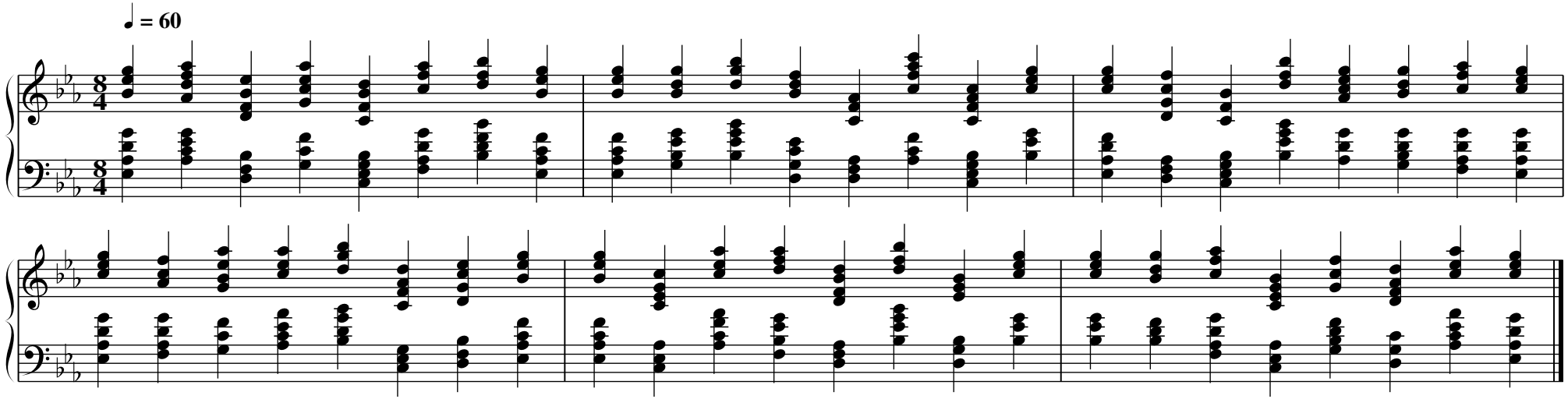
Trefoil in F



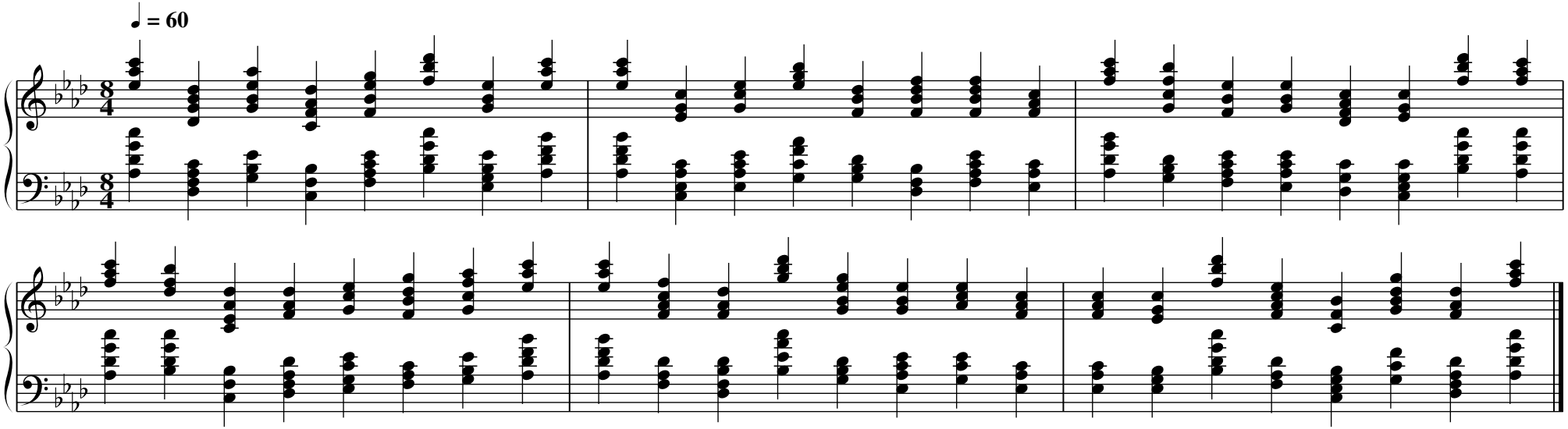
Trefoil in Bb



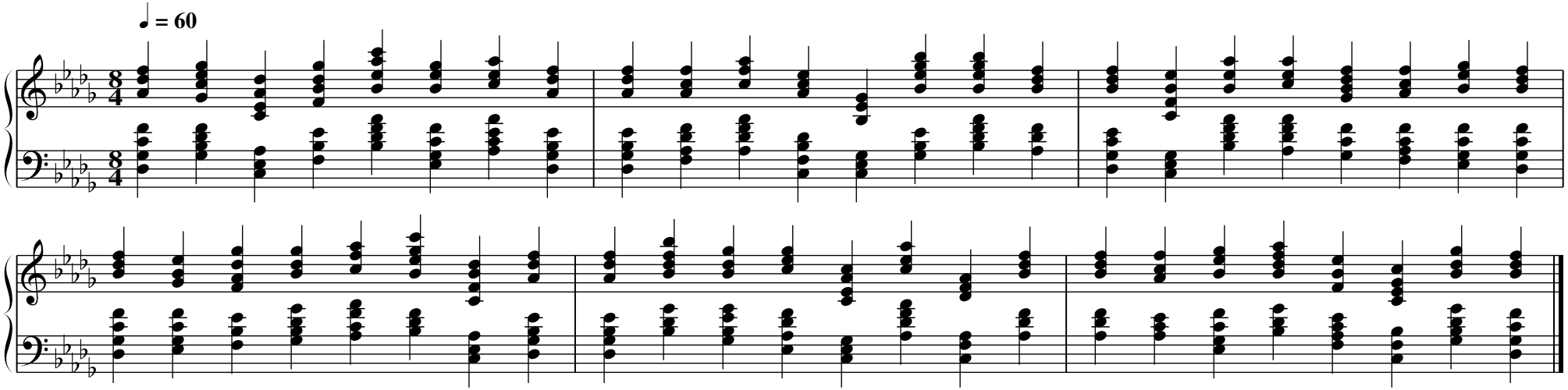
Trefoil in Eb



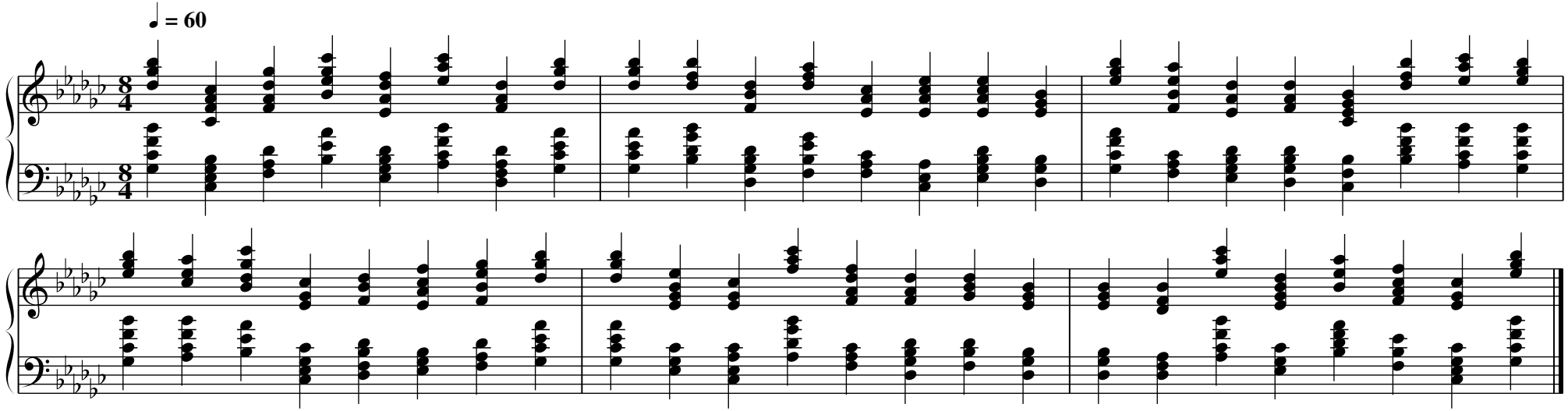
Trefoil in Ab



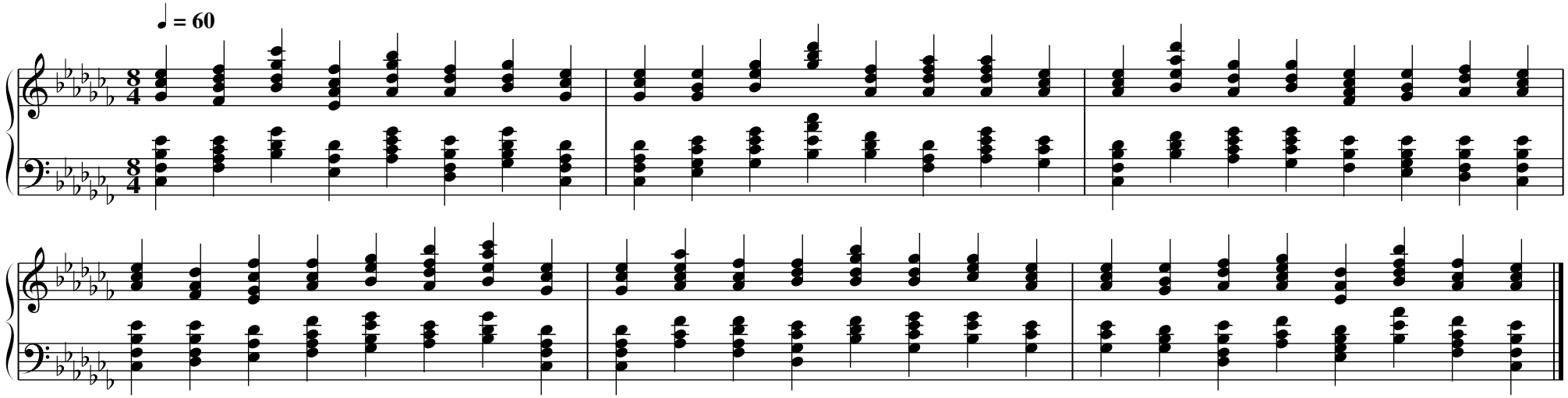
Trefoil in Db



Trefoil in Gb



Trefoil in Cb



Deg	LH	G	RH	Roman	Roman E♭	B♭	F	C	G	D	A
					E♭	B♭	F	C	G	D	A
CCW4											
I	333	**1**	32	IV△s^{2} I^{2}	E,A,DG Beg	B,EAd fbd’	F,B,EA cfa	C,F,B,E Gce	G,CFB dgb	D,G,CF Adf	A,DGc eac’
IV	222	**0**	322	IV△ vii°^{2}+8	A,CEG Adfa	E,G,B,D EAce	B,DFA Begb	F,A,CE FBdf	C,E,G,B, CFAc	G,B,DF Gceg	D,F,A,C DGBd
vii°	23	**1**	233	V^{1} I△s2^{3}	D,F,B, DFBe	A,CF Acfb	E,G,C EGcf	B,DG Bdgc’	F,A,D FAdg	C,E,A, CEAd	G,B,E GBea
iii	33	**0**	323	vis^{2} IV△^{3}	G,CF Gcea	D,G,C DGBe	A,DG Adfb	E,A,D EAcf	B,EA Begc’	F,B,E FBdg	C,F,B, CFAd
vi	222	**0**	332	vi7 V+2	C,E,G,B, CFBd	G,B,DF Gcfa	D,F,A,C DGce	A,CEG Adgb	E,G,B,D EAdf	B,DFA Beac’	F,A,CE FBeg
ii	233	**2**	32	iii7s2^{3} ii^{2}	F,A,DG cfa	C,E,A,D Gce	G,B,EA dgb	D,F,B,E Adf	A,CFB eac’	E,G,CF Beg	B,DGc fbd’
V	223	**1**	23	V+8 V^{1}	B,DFB dfb	F,A,CF Acf	C,E,G,C EGc	G,B,DG Bdg	D,F,A,D FAd	A,CEA cea	E,G,B,E GBe
I	323	**2**	32	ii7^{3} I^{2}	E,A,CF Beg	B,EGc fbd’	F,B,DG cfa	C,F,A,D Gce	G,CEA dgb	D,G,B,E Adf	A,DFB eac’
CW3											
I	323	**2**	32	ii7^{3} I^{2}	E,A,CF Beg	B,EGc fbd’	F,B,DG cfa	C,F,A,D Gce	G,CEA dgb	D,G,B,E Adf	A,DFB eac’
iii	232	**1**	23	I^{1}+8 iii^{1}	G,B,EG Bdg	D,F,B,D FAd	A,CFA cea	E,G,CE GBe	B,DGB dfb	F,A,DF Acf	C,E,A,C EGc
V	322	**1**	32	I^{2}+8 iii^{2}	B,EGB dgb	F,B,DF Adf	C,F,A,C EAc	G,CEG Beg	D,G,B,D FBd	A,DFA cfa	E,A,CE Gce
vii°	332	**3**	22	vi+2 V	D,G,CE Bdf	A,DGB fac’	E,A,DF ceg	B,EAc gbd’	F,B,EG dfa	C,F,B,D Ace	G,CFA egb
ii	22	**1**	32	vii° ii^{2}	D,F,A, CFA	A,CE Gce	E,G,B, DGB	B,DF Adf	F,A,C EAc	C,E,G, B,EG	G,B,D FBd
IV	23	**1**	322	ii^{1} ii^{2}+8	A,CF cfac’	E,G,C Gceg	B,DG dgbd’	F,A,D Adfa	C,E,A, EAce	G,B,E Begb	D,F,B, FBdf
vi	222	**0**	322	vi7 ii^{2}+8	C,E,G,B, CFAc	G,B,DF Gceg	D,F,A,C DGBd	A,CEG Adfa	E,G,B,D EAce	B,DFA Begb	F,A,CE FBdf
I	32	**2**	22	I^{2} vi	B,EG ceg	F,B,D GBd	C,F,A, DFA	G,CE Ace	D,G,B, EGB	A,DF Bdf	E,A,C FAc
CCW2											
I	332	**3**	22	I△9s-5 vi	E,A,DF ceg	B,EAc gbd’	F,B,EG dfa	C,F,B,D Ace	G,CFA egb	D,G,CE Bdf	A,DGB fac’
vii°	22	**2**	333	vii° iii7s^{2}	D,F,A, DGcf	A,CE Adgc’	E,G,B, EAdg	B,DF Bead’	F,A,C FBea	C,E,G, CFBe	G,B,D Gcfb
vi	222	**0**	33	vi7 Vs2^{1}	C,E,G,B, CFB	G,B,DF Gcf	D,F,A,C DGc	A,CEG Adg	E,G,B,D EAd	B,DFA Bea	F,A,CE FBe
V	322	**1**	23	I^{2}+8 V^{1}	B,EGB dfb	F,B,DF Acf	C,F,A,C EGc	G,CEG Bdg	D,G,B,D FAd	A,DFA cea	E,A,CE GBe
IV	33	**0**	222	iiis2^{1} IV△	A,DG Aceg	E,A,D EGBd	B,EA Bdfa	F,B,E FAce	C,F,B, CEGB	G,CF GBdf	D,G,C DFAc
iii	223	**1**	23	iii+8 iii^{1}	G,B,DG Bdg	D,F,A,D FAd	A,CEA cea	E,G,B,E GBe	B,DFB dfb	F,A,CF Acf	C,E,G,C EGc
ii	233	**2**	32	iii7s2^{3} ii^{2}	F,A,DG cfa	C,E,A,D Gce	G,B,EA dgb	D,F,B,E Adf	A,CFB eac’	E,G,CF Beg	B,DGc fbd’
I	333	**2**	22	IV△s^{2} vi	E,A,DG ceg	B,EAd gbd’	F,B,EA dfa	C,F,B,E Ace	G,CFB egb	D,G,CF Bdf	A,DGc fac’

Deg	LH	G	RH	Roman	E♭	B♭	F	C	G	D	A
CW2											
I	333	**2**	22	IVΔs^2 vi	E,A,DG ceg	B,EAd gbd’	F,B,EA dfa	C,F,B,E Ace	G,CFB egb	D,G,CF Bdf	A,DGc fac’
ii	233	**0**	23	iii7s2^3 ii^1	F,A,DG Acf	C,E,A,D EGc	G,B,EA Bdg	D,F,B,E FAd	A,CFB cea	E,G,CF GBe	B,DGc dfb
iii	33	**0**	233	vis^2 Is+3	G,CF GBea	D,G,C DFBe	A,DG Acfb	E,A,D EGcf	B,EA Bdgc’	F,B,E FAdg	C,F,B, CEAd
IV	223	**1**	23	IV+8 IV^1	A,CEA cea	E,G,B,E GBe	B,DFB dfb	F,A,CF Acf	C,E,G,C EGc	G,B,DG Bdg	D,F,A,D FAd
V	232	**1**	32	iii^1+8 iii^2	B,DGB dgb	F,A,DF Adf	C,E,A,C EAc	G,B,EG Beg	D,F,B,D FBd	A,CFA cfa	E,G,CE Gce
vi	22	**2**	323	vi viiø7^3	C,E,G, CFAd	G,B,D Gcea	D,F,A, DGBe	A,CE Adfb	E,G,B, EAcf	B,DF Begc’	F,A,C FBdg
vii°	23	**1**	332	V^1 vi+2	D,F,B, DGce	A,CF Adgb	E,G,C EAdf	B,DG Beac’	F,A,D FBeg	C,E,A, CFBd	G,B,E Gcfa
I	323	**2**	32	ii7^3 I^2	E,A,CF Beg	B,EGc fbd’	F,B,DG cfa	C,F,A,D Gce	G,CEA dgb	D,G,B,E Adf	A,DFB eac’
CCW3											
I	323	**2**	32	ii7^3 I^2	E,A,CF Beg	B,EGc fbd’	F,B,DG cfa	C,F,A,D Gce	G,CEA dgb	D,G,B,E Adf	A,DFB eac’
vi	23	**1**	223	IV^1 vi+8	C,E,A, CEGc	G,B,E GBdg	D,F,B, DFAd	A,CF Acea	E,G,C EGBe	B,DG Bdfb	F,A,D FAcf
IV	232	**1**	23	ii^1+8 IV^1	A,CFA cea	E,G,CE GBe	B,DGB dfb	F,A,DF Acf	C,E,A,C EGc	G,B,EG Bdg	D,F,B,D FAd
ii	332	**3**	22	I+2 vii°	F,B,EG dfa	C,F,B,D Ace	G,CFA egb	D,G,CE Bdf	A,DGB fac’	E,A,DF ceg	B,EAc gbd’
vii°	22	**2**	232	vii° V^1+8	D,F,A, DFBd	A,CE Acfa	E,G,B, EGce	B,DF Bdgb	F,A,C FAdf	C,E,G, CEAc	G,B,D GBeg
V	322	**1**	23	I^2+8 V^1	B,EGB dfb	F,B,DF Acf	C,F,A,C EGc	G,CEG Bdg	D,G,B,D FAd	A,DFA cea	E,A,CE GBe
iii	32	**2**	22	iii^2 I	D,G,B, EGB	A,DF Bdf	E,A,C FAc	B,EG ceg	F,B,D GBd	C,F,A, DFA	G,CE Ace
I	32	**2**	22	I^2 vi	B,EG ceg	F,B,D GBd	C,F,A, DFA	G,CE Ace	D,G,B, EGB	A,DF Bdf	E,A,C FAc
CW4											
I	32	**2**	22	I^2 vi	B,EG ceg	F,B,D GBd	C,F,A, DFA	G,CE Ace	D,G,B, EGB	A,DF Bdf	E,A,C FAc
V	22	**2**	23	V iii^1	B,DF Bdg	F,A,C FAd	C,E,G, CEA	G,B,D GBe	D,F,A, DFB	A,CE Acf	E,G,B, EGc
ii	233	**2**	32	iii7s2^3 ii^2	F,A,DG cfa	C,E,A,D Gce	G,B,EA dgb	D,F,B,E Adf	A,CFB eac’	E,G,CF Beg	B,DGc fbd’
vi	23	**1**	222	IV^1 vi7	C,E,A, CEGB	G,B,E GBdf	D,F,B, DFAc	A,CF Aceg	E,G,C EGBd	B,DG Bdfa	F,A,D FAce
iii	222	**0**	33	iii7 vis^2	G,B,DF Gcf	D,F,A,C DGc	A,CEG Adg	E,G,B,D EAd	B,DFA Bea	F,A,CE FBe	C,E,G,B, CFB
vii°	33	**0**	223	iiis^2 vii°+8	D,G,C DFAd	A,DG Acea	E,A,D EGBe	B,EA Bdfb	F,B,E FAcf	C,F,B, CEGc	G,CF GBdg
IV	223	**1**	23	IV+8 IV^1	A,CEA cea	E,G,B,E GBe	B,DFB dfb	F,A,CF Acf	C,E,G,C EGc	G,B,DG Bdg	D,F,A,D FAd
I	333	**2**	22	IVΔs^2 vi	E,A,DG ceg	B,EAd gbd’	F,B,EA dfa	C,F,B,E Ace	G,CFB egb	D,G,CF Bdf	A,DGc fac’

Trefoil Chord Reference — 42 Unique Voicings

Major / Dominant

Chord	Notes	ABC	Pat
I	CEG	[C,E,G,]	32
I^{2}	GCE	[G,CE]	32
I^{2}+8	GCEG	[G,CEG]	322
I^{1}+8	EGCE	[E,G,CE]	223
I+2	DGCE	[D,G,CE]	332
Is+3	EGCF	[E,G,CF]	233
I$\triangle$9s-5	CFBD	[C,F,B,D]	332
I$\triangle$s2^{3}	BDGC	[B,DGc]	233
IV^{1}	ACF	[A,CF]	23
IV+8	FACF	[F,A,CF]	223
IV$\triangle$	FACE	[F,A,CE]	222
IV$\triangle$s^{2}	CFBE	[C,F,B,E]	333
IV$\triangle^{3}$	EACF	[E,A,CF]	323
V	GBD	[G,B,D]	22
V^{1}	BDG	[B,DG]	23
V+8	GBDG	[G,B,DG]	223
V^{1}+8	BDGB	[B,DGB]	232
V+2	ADGB	[A,DGB]	332
Vs2^{1}	ADG	[A,DG]	33

Minor / Diminished

Chord	Notes	ABC	Pat
ii^{1}	FAD	[F,A,D]	23
ii^{2}	ADF	[A,DF]	23
ii^{1}+8	FADF	[F,A,DF]	232
ii^{2}+8	ADFA	[A,DFA]	323
ii7^{3}	CFAD	[C,F,A,D]	323
iii^{1}	GBE	[G,B,E]	23
iii^{2}	BEG	[B,EG]	32
iii+8	EGBE	[E,G,B,E]	223
iii^{1}+8	GBEG	[G,B,EG]	322
iii7	EGBD	[E,G,B,D]	222
iii7s^{2}	BEAD	[B,EAd]	332
iii7s2^{3}	DFBE	[D,F,B,E]	233
iiis^{2}	BEA	[B,EA]	33
iiis2^{1}	FBE	[F,B,E]	33
vi	ACE	[A,CE]	22
vi+2	BEAC	[B,EAc]	332
vi+8	ACEA	[A,CEA]	222
vi7	ACEG	[A,CEG]	222
vis^{2}	EAD	[E,A,D]	33
vii$^{\circ}$	BDF	[B,DF]	22
vii$^{\circ}$+8	BDFB	[B,DFB]	223
vii$^{\circ 2}$+8	FBDF	[F,B,DF]	332
viiø7^{3}	ADFB	[A,DFB]	233

ABC Note Inventory — Trefoil Voicing Range

The trefoil voicing system uses only diatonic notes in the range C,, to c′ (ABC notation). The melody voice may contain accidentals, but the RH and LH grand staff voices use **only** these notes — no sharps, no flats, no naturals. The key signature handles all accidentals automatically.

Octave	ABC Notes	Staff	Range
1	`C, D, E, F, G, A, B,`	Bass clef, below staff	Low register
2	`C D E F G A B`	Bass clef / Middle C area	Mid–low register
3	`c d e f g a b`	Treble clef	Mid–high register
4	`c' d'`	Treble clef, above staff	High register

Rules for hymn arrangement:

- **Melody voice** (top staff): may contain accidentals from the original hymn
- **RH voice** (treble clef): diatonic only — uses only the 23 notes above
- **LH voice** (bass clef): diatonic only — uses only the 23 notes above
- Key signature handles all flats/sharps — zero accidentals in RH/LH
- All 12 finger patterns (3-note and 4-note) appear in both hands
- Every LH/RH pair is asymmetric (different finger pattern per hand)

Complete ABC Note List (low to high):

`C, D, E, F, G, A, B, C D E F G A B c d e f g a b c' d'`

23 diatonic notes × 15 pedal harp keys = 345 possible note positions
42 unique voicings × 15 keys = 630 chord instances
37 unique asymmetric LH/RH pairs × 6 sections = 222 practice positions